

Homework

You must complete all problems. All text responses must be written by hand; you may use electronic mediums such as a tablet or write them on paper and take a picture of it. All writing must be neat, organized, and legible. Workflow section problems will require a plot from software such as excel or python, and your code must be uploaded separately. Submissions will be uploaded to Blackboard. You may work with others, but your final work must be your own.

Section 1: Theoretical Concepts

Question 1 (CA): What is the difference between structured and unstructured data? Come up with 3 examples of each using materials science data and explain why that data is either structured or unstructured.

Question 2 (CA, CE): I am a materials scientist that has accumulated a set of both structured and unstructured data over the past 2 years doing experimental characterization of metal microstructures. My data is mostly in the format of images but with corresponding structural properties for each microstructure. I have over 250,000 images, and 1 GB worth of structural properties in over 1,000 separate files. First, identify all unstructured and structured data. Next, identify how you would compile this data to share with a colleague that wants to perform machine learning analysis on mapping the microstructures to their corresponding structural properties.

Question 3 (CA): Consider two scenarios, (1) a machine learning model that maps a molecular structure to potential energy, and (2) a machine learning model that maps SEM images to defect concentrations. For each scenario, describe the features, in detail, that one would need to identify to create an accurate mapping between the material's structure and the corresponding property. Also, describe the types of symmetry operations one would need to ensure that the descriptors respect to map onto the corresponding property.

Section 2: Problem Solving

Question 4 (CA, CB): Machine-learning models are often used in materials science to predict a property from a set of inexpensive descriptors. However, a useful model requires more than simply calling a fitting function. The data must be separated into training and testing subsets, the model must be fit using only the training data, and performance must be evaluated on data that were not used to train the model.

For this problem, you are given a synthetic dataset representing 260 hypothetical inorganic materials. Each material has six physically motivated descriptors and a target value, the bulk modulus. The dataset is synthetic so that every student has access to the same clean dataset; the workflow is representative of a real materials-informatics regression problem.

Supplied file: data/materials_regression.csv

The columns are:

- material_id: unique identifier
- avg_atomic_number: average atomic number of the constituent elements
- avg_electronegativity: average electronegativity
- density_g_cm3: density in g/cm³
- cohesive_energy_eV_atom: cohesive energy in eV/atom
- atomic_volume_A3_atom: atomic volume in Å³/atom
- packing_fraction: dimensionless packing fraction
- bulk_modulus_GPa: target property

Your task

Write a Python program that performs the following workflow.

A. Load and inspect the data

1. Read the supplied CSV file.
2. Use the six numerical descriptor columns as the input feature matrix **X**.
3. Use bulk_modulus_GPa as the target **y**.
4. Report the number of materials and the number of input features.

B. Create training and testing datasets

Use train_test_split from scikit-learn to place **80% of the materials in the training set and 20% in the test set**. Use random_state=42 so that everyone obtains the same split.

Report the number of samples in each subset.

C. Train a regression model

Train a LinearRegression model using **only the training data**. Use the fitted model to predict the bulk modulus for both the training and test sets.

D. Evaluate the model

For both the training and test sets, calculate:

- Mean absolute error, MAE
- Coefficient of determination, R^2

Print the four values clearly.

E. Visualize the results

Create a predicted-versus-actual plot for the **test set**. Include:

- Actual bulk modulus on the x-axis
- Predicted bulk modulus on the y-axis
- A dashed or solid $y = x$ reference line
- Axis labels with units
- A descriptive title

Save the figure as `problem1_predicted_vs_actual.png`.

F. Interpret the workflow

In approximately **150–250 words**, answer the following:

1. Is the model substantially better on the training data than on the test data? What would a large difference between the two imply?
2. Is MAE or R^2 easier to interpret physically for this problem? Explain.
3. Based on the test results, would you consider this model useful as a rapid screening model for bulk modulus? Explain what additional evidence you would want before using it on real materials.

What to submit

1. Your Python script.
2. `problem1_predicted_vs_actual.png`.

3. Your 150–250 word interpretation.

Important: You are not graded on obtaining a perfect model. You are graded on constructing the workflow correctly and interpreting the result appropriately.

Question 5 (CA, CB): Machine-learning algorithms do not automatically understand the physical meaning or units of an input feature. A value of 500 in one column is not inherently more important than a value of 0.02 in another, but some algorithms use numerical distances and therefore can be dominated by features with large numerical ranges.

You are given a synthetic dataset of 300 hypothetical materials. The goal is to predict thermal conductivity using a k-nearest-neighbors regression model. The descriptors intentionally span very different numerical scales: for example, grain size may be hundreds of nanometers while defect fraction is much smaller than one.

Supplied file: data/materials_scaling_pca.csv

Do **not** use material_id or material_class as model input features. material_class is provided only to help you interpret the PCA visualization.

Your task

A. Prepare a common train/test split

Use the following seven features:

- avg_atomic_number
- density_g_cm3
- band_gap_eV
- defect_fraction
- porosity_percent
- grain_size_nm
- mean_bond_strength_eV

Use thermal_conductivity_W_mK as the target.

Create an 80/20 train/test split using random_state=42. **Use this exact same split for every model below.**

B. Model 1: KNN using the raw features

Train a KNeighborsRegressor(n_neighbors=7) directly on the unscaled feature values.

Calculate and report the test MAE and test R².

C. Model 2: KNN after standardization

Fit a StandardScaler using **only the training features**. Transform both the training and test data using that fitted scaler. Train the same KNeighborsRegressor(`n_neighbors=7`) on the standardized training data.

Calculate and report the test MAE and test R^2 .

Explain in 2–4 sentences why the scaler must be fit using only the training data.

D. Compare the two models

Create a small table containing the test MAE and R^2 from the raw-feature and standardized-feature models. Which workflow performs better? By approximately how much does the MAE change?

E. Examine the representation using PCA

Perform two separate two-component PCA analyses on **all input samples for visualization only**:

1. PCA using the raw feature values.
2. PCA using standardized feature values.

Create a figure with two separate plots or two saved figures showing PC1 versus PC2. Color the points by `material_class`. For each PCA calculation, print the fraction of variance explained by PC1 and PC2.

F. Interpret the result

In approximately **150–250 words**, answer:

1. Why does scaling have such a large effect on a distance-based model such as KNN?
2. Inspect the raw-feature PCA loadings. Which feature dominates PC1, and why is that unsurprising given the units/ranges in the dataset?
3. Is standardization always scientifically correct? Give one situation in which you might intentionally preserve, transform, or weight features differently rather than automatically standardizing everything.

What to submit

1. Your Python script.
2. A table of raw versus scaled KNN performance.
3. Your PCA visualization(s).
4. Your 150–250 word interpretation.